Lakshya JEE 2.0 2025

Maths

Relations & Functions

DPP: 1

Q1 The relation R defined on the set $A=\{1,2,3,4,5\}$ by $\mathrm{R}=\left\{(x, y):\left|x^2-y^2\right|<16\right\}$, then:
(A) $\mathrm{R}=\{(1,1),(2,1),(3,1),(4,1),(2,3)\}$
(B) $\mathrm{R}=\{(2,2),(3,2),(4,2),(2,4)\}$
(C) $\mathrm{R}=\{(3,3),(4,3),(5,4),(3,4)\}$
(D) None of these

Q2 If $\mathrm{A}=\{1,2,3\}$ and $\mathrm{R}_1=\{(1,2),(3,2),(1,3)\}$
$\mathrm{R}_2=\{(1,3),(3,6),(2,1),(1,2)\}$, then:
(A) R_1 is a relation and R_2 is not a relation on A
(B) R_1 and R_2 both are relation on A
(C) R_1 and R_2 both are not relation on A
(D) None of these

Q3 Define a relation R on set $\mathrm{A}=\{2,3,5,6,10\}$ as $x \mathrm{R} y$ if '$x<y$ and x divides y', then the domain of relation is :
(A) $\{2,3,5,6,10\}$
(B) $\{2,3,5\}$
(C) $\{2,3,5,6\}$
(D) $\{2,5,10\}$

Q4 A relation R is defined from $\{2,3,4,5\}$ to $\{3,6,7,10\}$ by $x \mathrm{R} y \Leftrightarrow x$ is relatively prime to y. Then domain of R is:
(A) $\{2,3,5\}$
(B) $\{3,5\}$
(C) $\{2,3,4\}$
(D) $\{2,3,4,5\}$

Q5 Let R be a relation in N defined by $R=\left\{\left(1+x, 1+x^2\right): x \leq 5, x \in N\right\}$.
Which of the following is false?
(A) $R=\{(2,2),(3,5),(4,10),(5,17),(6,25)\}$
(B) Domain of $R=\{2,3,4,5,6\}$
(C) Range of $R=\{2,5,10,17,26\}$
(D) None of these

Q6 The domain and range of the relation R given by $\mathrm{R}=\left\{(x, y): y=x+\frac{6}{x}\right.$; where $x, y \in N$ and $\left.x<6\right\}$, respectively, is:
(A) $\{1,2,3\},\{7,5\}$
(B) $\{1,2\},\{7,5\}$
(C) $\{2,3\},\{5\}$
(D) $\{2,3\},\{5,3\}$

Q7 Consider the following with regard to a relation R on a set of real numbers defined by $x \mathrm{R} y$ if and only if $3x+4y=5$. Consider the following three statements:
(1) $0 \mathrm{R} 1$ (2) $1 \mathrm{R} \frac{1}{2}$ (3) $\frac{2}{3} R \frac{3}{4}$
Which of the above are correct?
(A) 1 and 2 only (B) 1 and 3 only
(C) 2 and 3 only (D) 1,2 and 3

Q8 If $\mathrm{R}=\left\{(x, y) \mid x, y \in Z, x^2+y^2 \leq 4\right\}$ is a relation on Z, then domain of R is:
(A) $\{0,1,2\}$
(B) $\{0,-1,-2\}$
(C) $\{-2,-1,0,1,2\}$
(D) None of these

Q9 Let A be the set of first ten natural numbers and let R be a relation on A defined by $(x, y) \in R \Leftrightarrow x+2y=10$, i.e.,

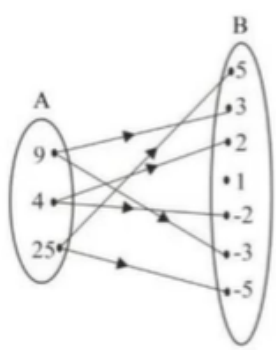
$\mathrm{R}=\{(x, y): x \in A, y \in A$ and $x+2y=10\}$. Then the domain and range of R is

(A) $\{2,4,6,8\},\{4,3,2,1\}$ respectively
(B) $\{4,3,2,1\},\{2,4,6,8\}$ respectively
(C) $\{1,2,3,4\},\{1,2,3,4\}$ respectively
(D) None of these

Q10 Define relation R_1 and R_2 on set $\mathrm{A}=\{2,3,5,7,10\}$ as $x\mathrm{R}_1 y$ if $x|(y-1)$ and $x\mathrm{R}_2 y$ if $x+y=10$ then the relation R given by $\mathrm{R}=\mathrm{R}_1 \cap \mathrm{R}_2$ is

(A) $\{(2,4)\}$
(B) $\{(3,7)\}$
(C) $\{(3,7),(5,5)\}$
(D) None of these

Q11 The figure given below shows a relation R between the sets A and B.



Then which of the following is correct?

I. The relation R in set builder form is $\{(x, y): \mathrm{x}$ is the square of $y, x \in A, y \in B\}$.

II. The domain of the relation R is $\{4,9,25\}$

III. The range of the relation R is $\{-5,-3,-2,2,3,5\}$

(A) Only I and II are true
(B) Only II and III are true
(C) I, II and III are true
(D) Neither I, II nor III are true

Q12 The relation R_1 and R_2 are defined from R to R as given below (R stand for set of real numbers)

$$R_1=\{(x, y):|x-3| \leq 1,|y-3| \leq 1\} \text{ and } R_2 = \left\{(x, y): 4x^2+9y^2-32x-54y+109 \leq 0\right\}$$

Choose the correct option

(A) $\mathrm{R}_1 \subset \mathrm{R}_2$
(B) $\mathrm{R}_1=\mathrm{R}_2$
(C) $\mathrm{R}_2 \subset \mathrm{R}_1$
(D) None of these

Q13 List all the relations on the set $\mathrm{A}=\{-5,5\}$.

Q14 Let $\mathrm{A}=\{2,3,4,5\}$ then how many relations can be defined on set A ?

(A) $(16)^2$
(B) $(2)^{14}$
(C) $(16)^4$
(D) $(4)^{16}$

Q15 For any two real numbers a and b, we define aRb if and only if $\sin^2 a+\cos^2 b=1$. The relation R is:

(A) Reflexive but not symmetric
(B) Symmetric but not transitive
(C) Transitive but not reflexive
(D) An equivalence relation.

Q16 Let S be the set of all real numbers. Then the relation $\mathrm{R}=\{(a, b): 1+ab>0\}$ on S is:

(A) Reflexive and symmetric but not transitive
(B) Reflexive, transitive and symmetric
(C) Reflexive and transitive but not symmetric.
(D) None of these

Q17 Let R be a relation defined on Q as follows: $a, b \in \mathrm{Q}, \mathrm{aRb}$ if and only if $|a-b| \leq 1$. Then which of the following is true?

(A) R is reflexive and symmetric
(B) R is reflexive and transitive
(C) R is symmetric only
(D) R is anti-symmetric only

Answer Key

Q1 (D)

Q2 (A)

Q3 (B)

Q4 (D)

Q5 (A)

Q6 (A)

Q7 (C)

Q8 (C)

Q9 (A)

Q10 (B)

Q11 (C)

Q12 (A)

Q13 $\phi, \{(-5,-5)\}, \{(-5,5)\}, \{(5,-5)\}, \{(5,5)\}, \{(-5,-5),(-5,5)\}, \{(-5,-5),(5,-5)\}, \{(-5,-5),(5,5)\}, \{(-5,5),(5,-5)\}, \{(-5,5),(5,5)\}, \{(5,-5),(5,5)\}, \{(-5,-5),(-5,5),(5,-5)\}, \{(-5,-5),(-5,5),(5,5)\}, \{(-5,5),(5,-5),(5,5)\}, \{(-5,-5),(5,-5),(5,5)\}, \{(-5,-5),(-5,5),(5,-5),(5,5)\}$

Q14 (C)

Q15 (D)

Q16 (A)

Q17 (A)

Android App | iOS App | PW Website